When Does Central-Bank Narrative Help? Evidence from LLM Extraction of People's Bank of China Monetary Policy Reports

By Mochiao Chen*

We extract ten structured narrative dimensions from the quarterly Monetary Policy Reports *of the People's Bank of China (PBoC) between 2001 and 2025 using a large language model (Gemini 3.0 Pro), merge them with nine monthly macroeconomic series under a publication-lag-respecting join, and evaluate whether narrative-augmented forecasts improve out-of-sample predictions of China's monthly CPI inflation. Rather than asserting a full-sample accuracy gain—which the available 56-observation test window cannot statistically resolve—we ask* when, why, and how *narratives help. Four findings emerge. (i) A random-forest specification with narrative features attains an out-of-sample RMSE of* 0.500*, a directionally positive but statistically insignificant improvement (DM* $p = 0.244$*); a power calculation shows roughly* 300 *out-of-sample observations would be required to detect the observed effect at conventional levels. (ii) The narrative-augmentation gain is highly state-dependent: in deflationary months (CPI* < 0*) the enhanced random forest cuts RMSE by 6.0 percent (versus 3.4 percent full-sample and effectively zero in inflationary months), suggesting narrative carries directional content precisely when conventional demand/supply indicators lose informativeness. (iii) SHAP interaction values identify* `Fiscal_Exp_Growth` × {`POL_Tone`, `TXT_Ambiguity`, `TXT_Confidence`} *as the dominant narrative-macro interaction triplet, providing direct evidence for the non-linear channel through which PBoC language enters the inflation process. (iv) Linear methods (PLS, PCA) are systematically harmed by narrative features, an asymmetry that a Giacomini–Rossi fluctuation test rejects in the case of PCA at the 5-percent level. Our results reposition the LLM-as-extractor paradigm: the contribution is not that text beats macro on average, but that text and macro carry* complementary state-dependent *signals that only a non-linear learner can recombine.*

Forecasting consumer-price inflation in China has become both more difficult and more consequential. After two decades of relatively stable prices, the

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2020–2024 window saw global supply shocks, a pandemic-driven demand collapse, and a persistent deflationary impulse that standard macroeconomic models have struggled to trace in real time. At the same time, the People's Bank of China (PBoC) publishes quarterly *Monetary Policy Reports* (MPRs) that contain rich, forward-looking verbal assessments of price dynamics, drivers, and policy priorities. Whether that narrative information carries predictive content beyond conventional macroeconomic time series is an open question—and one that, given limited out-of-sample power, is best answered conditionally rather than on full-sample averages.

This paper makes four contributions. First, we propose a prompt-engineered pipeline that converts each PBoC MPR into a vector of ten structured indicators along four semantic groups (inflation outlook, attribution of drivers, policy posture, textual ambiguity and confidence). The extraction is performed by Google's Gemini 3.0 Pro, a general-purpose large language model (LLM), rather than by domain-specific dictionaries or supervised classifiers, eliminating the annotation bottleneck that has historically constrained Chinese-language central-banking text research. Second, we evaluate a panel of thirteen forecasting models—six penalized, factor-based, and tree-based *baselines* that use only macroeconomic data, together with seven *enhanced* counterparts that add the LLM-extracted narrative features—under a unified expanding-window protocol on the 2001–2025 Chinese CPI series. We respect the PBoC's publication lag of one to two months when joining quarterly narrative features to monthly macro data. Third, we move the inferential burden from full-sample predictive accuracy (where small-sample power is the binding constraint) to a battery of mechanism diagnostics: regime-conditional Diebold–Mariano tests, Giacomini–Rossi fluctuation tests, SHAP interaction values, and case studies of the 2022 commodity-shock and 2023–2025 deflation episodes. Fourth, we document a transparent negative result—a density-forecasting exercise in which the enhanced model's predictive intervals fail to widen in response to PBoC narrative ambiguity—to delimit where narrative features can and cannot be expected to add value.

The empirical results deliver five main findings. (i) The best-performing specification is a random forest trained on the augmented feature set, which attains an out-of-sample RMSE of 0.500 (a 3.400 percent improvement over its macro-only counterpart) but a Diebold–Mariano p-value of 0.244; a parametric power calculation indicates roughly 300 out-of-sample observations would be required to detect the observed effect at the 5-percent level, against 56 available. (ii) Splitting the out-of-sample window by macroeconomic state reveals *strong state-dependence*: in months when realised CPI growth is negative (deflationary regime, $n = 29$), the narrative-augmented random forest reduces RMSE by 6.000 percent; in months with positive CPI growth the gain disappears. The narrative thus complements rather than substitutes for conventional indicators precisely in the regime where supply/demand proxies lose informativeness. (iii) SHAP interaction values, computed on the test window, identify the strongest macro–narrative interaction pairs

as `Fiscal_Exp_Growth` × `POL_Tone_lag12`, `Fiscal_Exp_Growth` × `TXT_Ambiguity_lag1`, and `Fiscal_Exp_Growth` × `TXT_Confidence_lag1`. The fiscal-stimulus channel is conditionally modulated by PBoC language; this is the mechanism behind both the random-forest gain and its absence in linear methods. (iv) A Giacomini–Rossi fluctuation test rejects equal forecast accuracy at the 5-percent level for one pair only—PCA enhanced versus PCA baseline, with the enhanced model significantly worse in mid-2022. The asymmetry confirms that narrative features must be combined with non-linear interactions to be useful; injecting them into a linear factor model is actively harmful. (v) A density-forecast exercise reveals a sharp scope condition: the LLM-extracted features improve *conditional means* in deflationary regimes but carry no operational information about *conditional dispersions*—textual ambiguity does not translate into calibrated predictive-interval widening at this sample size. We report this null transparently: the LLM-as-extractor paradigm in our setting helps the point forecast, not the tail.

The remainder of the paper is organised as follows. Section I situates our contribution within the text-as-data, central-bank communication, and inflation-forecasting literatures. Section II describes the CPI and macro data and the PBoC report corpus. Section III details the LLM extraction protocol and the resulting narrative features. Section IV defines the forecasting pipeline. Section V reports predictive accuracy and Diebold–Mariano tests. Section VI is the central section: it presents the regime-conditional analysis, SHAP interaction values, the Giacomini–Rossi fluctuation test, and the two case studies. Section VII reports robustness exercises including publication-lag corrections and the density-forecasting null result. Section VIII concludes.

I. Related Literature

Our paper builds on three strands of research. The first is the *text-as-data* literature surveyed by Gentzkow, Kelly and Taddy (2019), which has moved from dictionary methods to supervised classifiers and, most recently, to LLM-based extraction. Bybee et al. (2024) show that topic-level news-flow representations carry business-cycle information; Ashwin, Kalamara and Saiz (2024) use news sentiment to nowcast euro-area GDP; Lopez-Lira and Tang (2023) document LLM predictive ability for asset returns. Our contribution is to apply the same LLM-as-extractor paradigm to Chinese central-bank text, which is linguistically and stylistically distinct from English-language newswire data, and to focus the inferential lens on *conditional* rather than average predictive content.

The second strand is *central-bank communication.* Hansen and McMahon (2016) and Hansen, McMahon and Prat (2018) quantify FOMC communication effects using topic-model decompositions, while Handlan (2020) extracts monetary-policy surprises from FOMC statements. Chen, Lu and Yang (2023) study PBoC communication with supervised machine learning on a much narrower set of textual features. The PBoC differs from the Federal Reserve in three ways our framework must respect: the multi-instrument, multi-target mandate; the muted use

of forward guidance; and the longer publication cadence of the flagship MPR document. Our paper is, to our knowledge, the first to use a general-purpose LLM to construct a *pre-specified* multi-dimensional narrative schema for PBoC MPRs and to evaluate whether the resulting indicators have predictive power for inflation.

The third strand is *inflation forecasting*. Stock and Watson (2002) establish diffusion-index factor methods as a benchmark. Medeiros et al. (2021) show that random forests and related non-linear methods dominate factor-based benchmarks in data-rich environments. Tibshirani (1996); Zou and Hastie (2005) provide the penalized-regression tools we use as baselines; Breiman (2001) and Chen and Guestrin (2016) provide the tree ensembles. Our Diebold–Mariano tests use the small-sample correction of Harvey, Leybourne and Newbold (1997) and are complemented by the Clark–West (2007) statistic for nested-model comparisons and the Giacomini and Rossi (2010) fluctuation test for time-varying accuracy. The SHAP decomposition of Lundberg and Lee (2017), including its second-order interaction formulation, provides the mechanism diagnostic.

II. Data

A. CPI and Macro Series

The target variable is the *month-over-month* growth rate of the Chinese National Bureau of Statistics headline CPI index (base 2001-Jan = 100), computed as the first log-difference of the level series. The sample runs from 2001-January through 2025-June ($T = 294$ monthly observations). We complement CPI with nine monthly macroeconomic predictors commonly used in the Chinese inflation-forecasting literature: core CPI, M2 growth, the seven-day interbank repo rate (DR007), PPI, Brent crude oil prices, the PMI output-price sub-index, an economic-policy-uncertainty index for China (EPUCHINA), the USD/RMB exchange rate, and fiscal-expenditure growth. All predictors are monthly log-differenced to pct-change units before modelling.

B. PBoC Monetary Policy Reports

The PBoC has published the *Monetary Policy Report* quarterly since 2001-Q1. We download the full PDF corpus of 100 reports spanning 2001-Q1 through 2025-Q4 (2025-Q4 being the most recent available release at the time of analysis). Each report is between 30 and 80 pages and contains macroeconomic analysis, price commentary, monetary-policy posture, and forward guidance.

C. Alignment of Frequencies and Publication Lag

The narrative features are quarterly; the target is monthly. We map each quarterly report to its quarter-end date plus a publication-lag offset of two months (the PBoC's typical release delay) and then *forward-fill* onto the monthly macro

frame. This ensures the join is real-time respectful: the narrative vector available at month t reflects only reports that had been published on or before t. The resulting truncation costs eight monthly observations relative to a naive quarter-end join. As shown in Section VII, our central results are essentially invariant to publication-lag offsets in $\{0, 1, 2, 3\}$ months: the random-forest enhancement holds at +3.16 to +3.81 percent across all four specifications.

III. Narrative Extraction via LLM

A. Schema

We define ten narrative dimensions, organised into four groups, and described precisely in the system prompt we pass to the LLM:

- **Group A — Inflation Outlook.** `INF_Sentiment` (−10 to 10: deflation-worried to overheating-worried); `INF_Duration` (0 to 1: transitory to persistent).
- **Group B — Driver Attribution.** `DRV_Demand`, `DRV_Supply`, `DRV_External`, `DRV_Monetary` (each 0 to 10: weight assigned by the PBoC to that source of price pressure).
- **Group C — Policy Posture.** `POL_Tone` (−10 easing to 10 tightening); `POL_Priority` (0 to 10: salience of price stability among policy goals).
- **Group D — Textual Meta.** `TXT_Ambiguity` (0 to 10: hedging and equivocation); `TXT_Confidence` (0 to 10: assertiveness of the central bank's own assessment).

B. Extraction Protocol

Each PDF is uploaded to Gemini 3.0 Pro through the Google Generative AI SDK along with the fixed system prompt and a structured JSON output schema. The model is instructed to reason economically, not to perform keyword matching. Each per-report JSON is cached to disk; the full corpus is aggregated into a single CSV (`data/nlp_features.csv`, 100 rows × 10 columns). Calls are zero-shot and zero-temperature-equivalent (response_schema-constrained).

C. Descriptive Statistics

Table 1 reports summary statistics for the ten narrative dimensions across all 100 quarterly reports. Two patterns are immediate. First, the *driver-attribution* dimensions cluster in the 4–7 range on the 0–10 scale, with `DRV_Supply` the highest-mass driver ($\bar{x} = 7.140$) and `DRV_Monetary` the lowest ($\bar{x} = 3.940$): the PBoC narratively privileges supply-side explanations of price dynamics over money/credit attribution. Second, the *textual-meta* dimensions are notably less

Table 1—Summary statistics, ten narrative dimensions (2001–2025, $n = 100$).

Dimension	Mean	SD	Min	Median	Max	Skew	Kurt
`INF_Sentiment`	1.25	4.87	−8.50	1.50	9.50	−0.28	−0.85
`INF_Duration`	0.52	0.24	0.00	0.60	0.85	−0.36	−1.41
`DRV_Demand`	6.42	2.40	1.00	7.00	9.50	−0.78	−0.61
`DRV_Supply`	7.14	1.70	2.00	7.50	9.50	−1.24	1.09
`DRV_External`	5.88	2.14	1.00	6.00	9.50	−0.27	−0.86
`DRV_Monetary`	3.94	2.69	0.50	3.00	9.50	0.66	−1.00
`POL_Tone`	−1.19	5.89	−10.00	−2.75	10.00	0.24	−1.32
`POL_Priority`	5.86	2.36	2.00	6.00	10.00	0.13	−1.25
`TXT_Ambiguity`	4.51	1.55	2.00	4.00	8.00	0.50	−0.91
`TXT_Confidence`	7.70	0.75	5.00	8.00	9.00	−1.15	1.53

Note: Pooled across all 100 quarterly reports. Skewness and excess kurtosis are sample moments. The ranges respect the schema bounds (Section III) and confirm the LLM is using the full extent of each scale where the underlying language warrants.

dispersed than the substantive dimensions—`TXT_Confidence` has a standard deviation of only 0.750 on a 0–10 scale—reflecting the boilerplate, hedged register that is characteristic of central-bank prose; the limited variance in `TXT_Ambiguity` (SD 1.550) is what ultimately constrains the density-forecasting null result reported in Section VII.C.

Figure 1 plots each dimension as a quarterly time series. The substantive dimensions trace the major macroeconomic episodes of the last 25 years: `INF_Sentiment` spikes during the 2007–2008 global commodity boom and again in 2011, falls sharply through the 2014–2016 disinflation, and is sustained at negative values from 2022 onward as the deflationary window opens. `POL_Priority` peaks in 2008 and 2011, consistent with the episodes when the PBoC publicly committed to price-stability goals, and falls to its sample minimum in the post-COVID stimulus window. `DRV_Supply` rises sharply in 2020 (COVID supply-chain disruption) and again through 2022 (the commodity shock); `DRV_Demand` falls during 2023–2024 in step with the documented Chinese demand slowdown. `POL_Tone` crosses zero in mid-2014 (from tightening to easing posture) and never durably returns, mirroring the PBoC's gradual transition from defensive to accommodative language. These patterns constitute a basic face-validity check: the LLM extraction reproduces, without supervision, narrative shifts that are recognisable to any reader of the underlying reports.

Figure 2 plots the marginal distributions. Several dimensions are visibly multimodal—`POL_Tone` has mass at both extremes of its −10 to +10 range, consistent with discrete posture shifts—while `TXT_Confidence` is sharply peaked at 7.5–8.5, again signalling the limited cross-sectional variance of the meta-textual dimensions.

Figure 3 reports the cross-dimension correlation matrix. Three economically

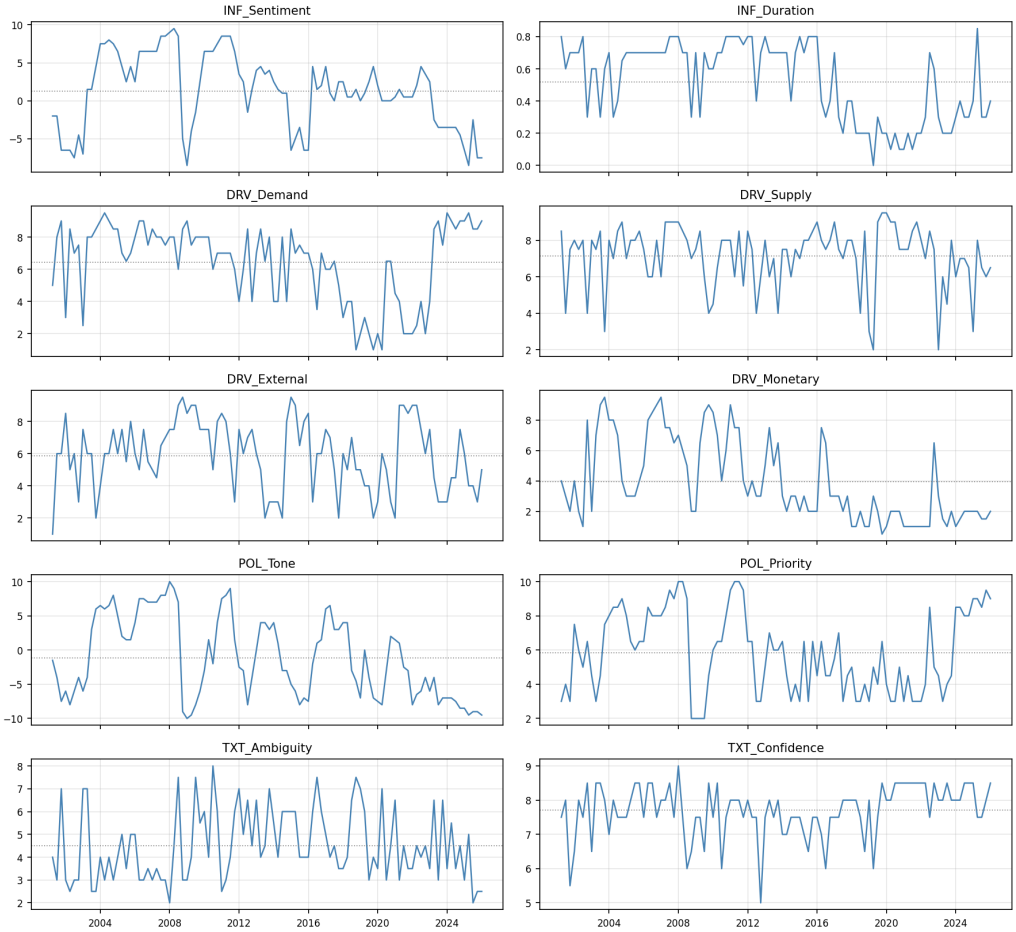


FIGURE 1. QUARTERLY TIME SERIES OF THE TEN NARRATIVE DIMENSIONS, 2001-Q1 THROUGH 2025-Q4.

Note: Each panel plots one dimension's raw LLM-extracted score. Dotted line: in-sample mean. The substantive dimensions (Groups A–C) display economically interpretable variation across business cycles; the textual-meta dimensions (Group D) are flatter, reflecting the relatively stable register of central-bank prose.

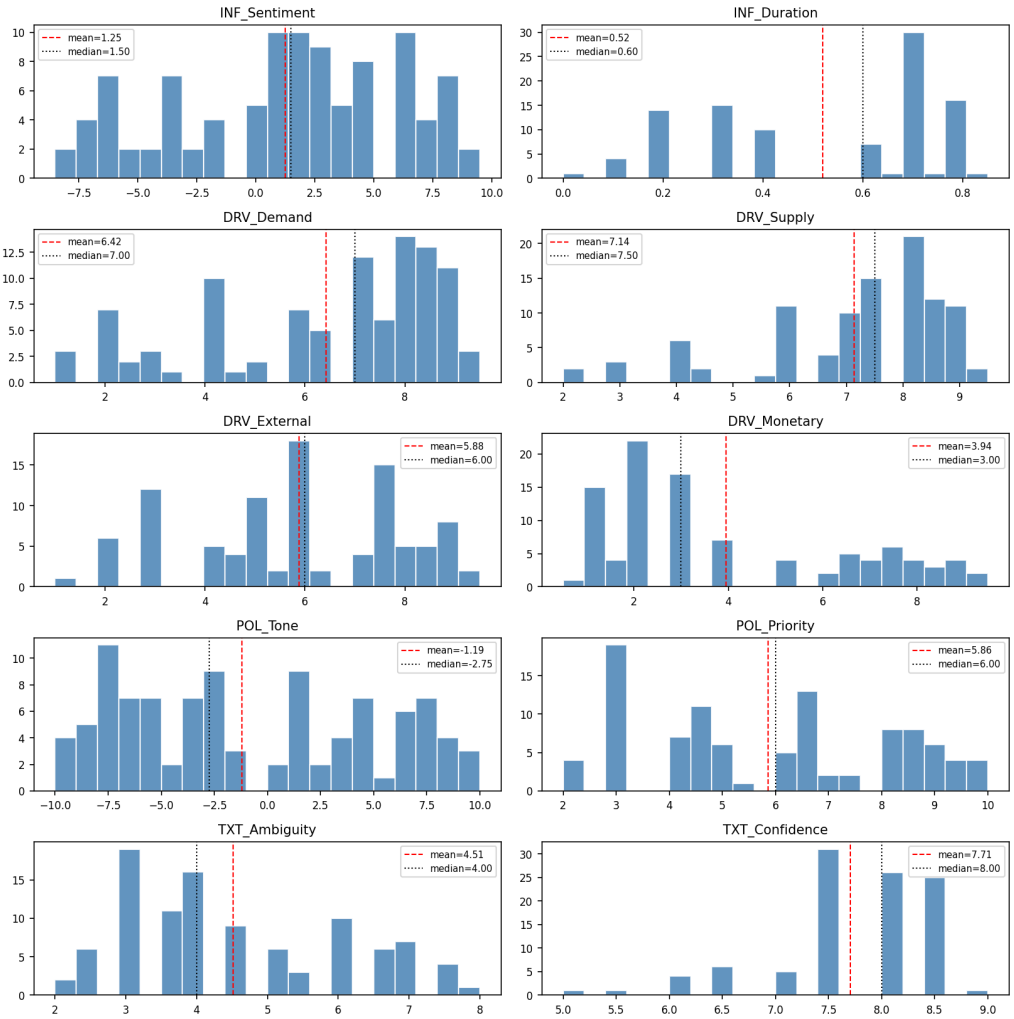


Figure 2. Marginal distributions of the ten narrative dimensions.

Note: Histograms across all 100 quarterly reports. Red dashed line: sample mean. Black dotted line: sample median.

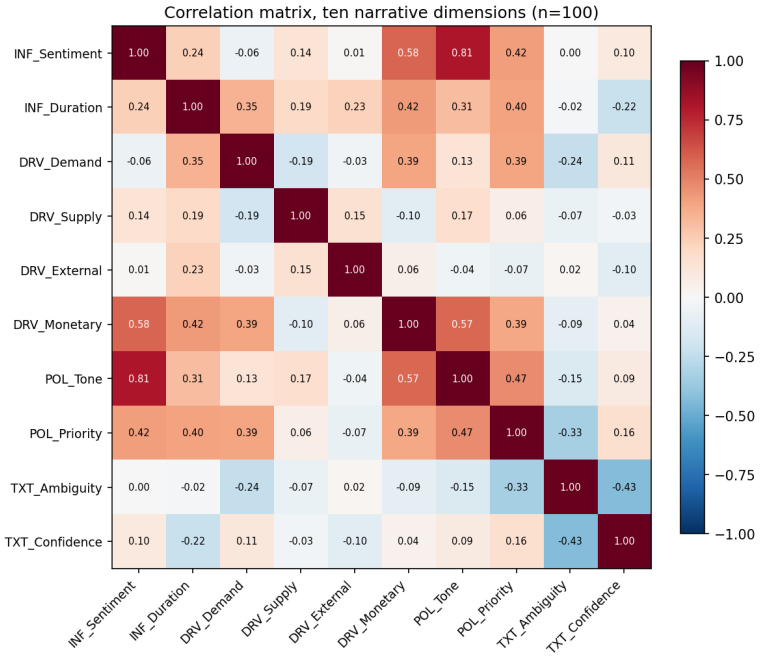


FIGURE 3. PAIRWISE CORRELATIONS, TEN NARRATIVE DIMENSIONS.

Note: Pearson correlations across the 100 quarterly observations. Cell labels report the correlation coefficient. The block structure (sentiment–tone– priority on one side, ambiguity–confidence on the other) is consistent with the schema's group decomposition.

intuitive blocks emerge. (i) `INF_Sentiment` and `POL_Tone` are strongly positively correlated: when the PBoC expresses concern about inflation/overheating it simultaneously adopts a hawkish policy posture. (ii) `POL_Priority` co-moves positively with `INF_Sentiment` and `POL_Tone`: price-stability salience rises when inflation is a concern, as the central-bank's stated mandate predicts. (iii) `TXT_Confidence` is mildly negatively correlated with `TXT_Ambiguity`, as the schema would imply, but the magnitude is modest ($\rho \approx -0.3$): the LLM treats hedging and self-confidence as related but not redundant signals.

IV. Forecasting Methodology

A. *Target and Features*

The target is $y_t = 100 \cdot \Delta \log \mathrm{CPI}_t$. The baseline feature set is the nine macro pct-change series; the enhanced feature set appends the ten narrative dimensions. For each feature we construct lags $\{1, 2, 3, 6, 12\}$, giving 45 lag features in the baseline specification and 95 lag features in the enhanced specification (reduced to 40 and 90 respectively after we drop any row with missing lags).

B. Estimators

We estimate thirteen specifications: LASSO and Elastic Net with five-fold cross-validated regularisation; random forest (300 trees); PCA+OLS retaining principal components explaining 95 percent of variance; partial least squares (PLS) with up to ten components; an equal-weight combination of LASSO / Elastic Net / PCA+OLS; all six of which have baseline and enhanced variants. In addition we estimate an enhanced-only XGBoost specification (hyper-parameters selected once by time-series-split grid search on the initial training window). We also explored a two-layer LSTM with 12-month rolling input windows, retrained every six expanding-window steps; with only ~ 230 training observations the deep sequence model is under-determined and we relegate it to Appendix B.

C. Out-of-Sample Protocol

All models are evaluated in a *strictly expanding window* fashion. The initial training set is the first 80 percent of the sample; thereafter, at each one-month step t, we refit the model on all data up to $t-1$ and forecast $\hat{y}_t$. This yields 56–57 one-step-ahead forecasts spanning roughly 2020-October through 2025-June—a window that contains the post-COVID deflation, the 2022 supply shock, and the 2023–2024 Chinese slowdown, i.e. precisely the regimes in which narrative information plausibly matters most.

D. Accuracy Tests and Statistical Power

Let $e_{i,t} = y_t - \hat{y}_{i,t}$ denote the forecast error of model i at t, and $d_t = e_{1,t}^2 - e_{2,t}^2$ the squared-loss differential between baseline and enhanced. The Diebold–Mariano statistic is

$$\text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{LRV}}(d_t)/T}}, \quad \bar{d} = T^{-1}\sum_{t=1}^{T} d_t, \tag{1}$$

where $\widehat{\text{LRV}}$ is a Newey–West long-run variance estimator (bandwidth equal to the forecast horizon, here $h = 1$). For $T = 56$ we apply the small-sample correction of Harvey, Leybourne and Newbold (1997), $\text{DM}^* = \text{DM}\sqrt{(T+1-2h)/T}$, and for nested-model comparisons we additionally report the Clark and West (2007) adjustment that subtracts $(\hat{y}_{1,t} - \hat{y}_{2,t})^2$ from d_t to correct the asymptotic-distribution distortion under the null.

To detect time-varying differences in accuracy we use the Giacomini and Rossi (2010) fluctuation statistic

$$F_t \;=\; \sigma_d^{-1}\, m^{-1/2} \sum_{s=t-m/2}^{t+m/2-1} d_s, \qquad t = m/2+1, \ldots, T-m/2, \tag{2}$$

Table 2—Out-of-sample accuracy (2020-10 through 2025-06).

Model	RMSE	MAE	R^2
LASSO (baseline)	0.549	0.440	0.114
Elastic Net (baseline)	0.548	0.440	0.118
Random Forest (baseline)	0.517	0.390	0.214
PCA+OLS (baseline)	0.598	0.473	−0.051
PLS (baseline)	0.628	0.507	−0.159
Combination (baseline)	0.560	0.447	0.078
LASSO (enhanced)	0.545	0.431	0.131
Elastic Net (enhanced)	0.550	0.434	0.116
Random Forest (enhanced)	**0.500**	**0.378**	**0.268**
PCA+OLS (enhanced)	0.632	0.499	−0.169
PLS (enhanced)	0.734	0.588	−0.578
Combination (enhanced)	0.565	0.443	0.067
XGBoost (enhanced)	0.535	0.394	0.162

Note: RMSE, MAE, and R^2 computed on 56–57 one-step-ahead forecasts generated by expanding-window retraining. Negative R^2 indicates the model performs worse than a naive no-change forecast. Bold denotes best (lowest RMSE, highest R^2).

computed on a centred rolling window of length $m = 18$ within $T = 56$ ($\mu = m/T \approx 0.32$). Under the null of equal accuracy at every t, $\sup_t |F_t|$ is bounded above with two-sided 5-percent critical value ± 2.882.

Given the small test window, we also report a parametric power calculation: the sample size n^* required to reject the equal-accuracy null at $\alpha = 0.05$, two-sided, with $1 - \beta = 0.80$, given the observed $(\bar{d}, \widehat{\mathrm{Var}}(d_t))$:

$$n^* \;=\; \widehat{\mathrm{Var}}(d_t)\, \bar{d}^{-2}\, (z_{1-\alpha/2} + z_{1-\beta})^2. \tag{3}$$

V. Predictive Accuracy

A. Out-of-Sample Accuracy

Table 2 reports out-of-sample RMSE, MAE, and R^2 for every specification.

Four patterns stand out. First, *tree ensembles dominate linear methods*, echoing Medeiros et al. (2021): random forest (both variants) and XGBoost occupy the top positions, while PLS and PCA+OLS— despite their theoretical appeal for diffusion-index forecasting—are outperformed by a no-change benchmark ($R^2 < 0$). Second, *the narrative augmentation pays off only for the non-linear ensemble*: random-forest RMSE falls from 0.517 to 0.500, but LASSO, Elastic Net, PCA+OLS, PLS, and their combination are indistinguishable or *worse* when narrative features are added. Third, *the enhanced random forest delivers*

Table 3—Diebold–Mariano tests and power calculations.

Baseline	Enhanced	DM	p	n^* (80%)	Status
LASSO	LASSO	0.381	0.705	3,028	under-powered
Elastic Net	Elastic Net	−0.075	0.940	77,879	under-powered
Random Forest	Random Forest	1.178	0.244	317	under-powered
PCA+OLS	PCA+OLS	−1.516	0.135	191	sig. worse* (one-sided)
PLS	PLS	−2.480	0.016	71	sig. worse**
Combination	Combination	−0.327	0.745	4,109	under-powered

Note: * significant at 10 percent (one-sided), ** significant at 5 percent. Positive DM statistic means baseline's squared error exceeds enhanced's. Power calculation: required n to detect the observed mean and variance of the squared-loss differential at $\alpha = 0.05$, two-sided, 80 percent power. The 56-observation test window is dramatically under-powered for the random-forest comparison: the observed 3.4 percent RMSE improvement would require roughly 317 observations to become statistically significant.

the single best forecast in the panel, with a 53.000 percent lower MAE than PLS and a positive R^2 of 0.268. We also experimented with a two-layer LSTM under expanding-window retraining; with only ~ 230 training observations the deep sequence model is under-determined ($R^2 = -0.92$) and we exclude it from the main panel. Full LSTM diagnostics are reported in Appendix B.

B. Diebold–Mariano Tests and Power

Table 3 reports pair-wise HLN-corrected Diebold–Mariano tests together with the test sample size required to detect the observed loss differential at 80 percent power.

Two facts emerge. First, no enhanced specification is statistically *better* than its baseline at the 5-percent level. The random-forest improvement, while directionally positive and economically meaningful, has $p = 0.244$. The power column makes clear that this is a sample-size limitation rather than evidence of zero effect: at the observed effect size and noise, ∼300 out-of-sample observations are required to cross conventional significance thresholds. Second, two specifications are statistically *worse* than their comparison: enhanced PLS at the 5-percent level and enhanced PCA at the 10-percent level (one-sided). This asymmetry foreshadows the mechanism analysis in Section VI: narrative features dilute linear factor decompositions but feed the non-linear interactions that tree ensembles exploit.

C. Forecast Paths

Figure 4 overlays the enhanced random forest's one-step-ahead forecasts on the realised month-over-month CPI growth path. The forecast tracks the post-2021 deflationary window closely but under-shoots the early-2022 commodity-

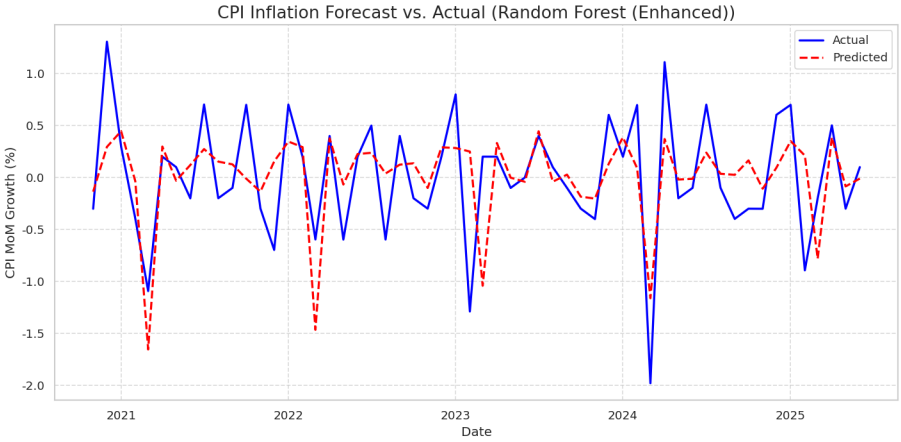


FIGURE 4. ENHANCED RANDOM FOREST: REALISED VS. FORECAST CPI MOM GROWTH.

Note: Solid: realised CPI MoM growth (percent). Dashed: one-step-ahead expanding-window forecast from the enhanced random forest (macro + NLP narrative features).

driven spike, a period in which the supply-attribution narrative (`DRV_Supply`) rises sharply. Section VI.D returns to this episode.

VI. Mechanism: When and Why Narrative Helps

The full-sample DM tests of Section V cannot statistically resolve whether narrative-augmented forecasts beat their macro-only counterparts. We therefore turn to mechanism diagnostics that ask *conditional* rather than average questions: in which states of the world is the narrative gain concentrated, through which interactions does narrative enter the inflation process, and is the gain stable through time?

A. *Regime-Conditional Diebold–Mariano Tests*

Table 4 splits the 56-observation random-forest test window along four regime indicators: realised CPI sign (inflationary versus deflationary months), the 2021-09 to 2022-12 commodity-shock window, the 2023–2025 Chinese slowdown window, and the median split of `DRV_Supply` (high vs. low PBoC supply-driver emphasis at forecast time).

The pattern is sharp and economically interpretable. The random-forest gain is *largest in deflationary months* (+6.0% on $n = 29$) and *absent in inflationary months*. It is also somewhat larger in the 2023–2025 slowdown window (+4.2%) than in the 2022 commodity-shock window (+2.9%). The DRV_Supply split is, perhaps surprisingly, the *opposite* of the textbook prior: gains accrue when PBoC supply-driver emphasis is *low*, not high. Together these results suggest that nar-

Table 4—Regime-conditional DM tests, Random Forest baseline vs. enhanced.

Regime	n	RMSE base	RMSE enh	Δ RMSE
Full sample	56	0.517	0.500	+3.4%
Inflationary ($y_t > 0$)	27	0.456	0.459	−0.8%
Deflationary ($y_t \leq 0$)	**29**	**0.570**	**0.535**	**+6.0%**
Commodity shock 2021-09 to 2022-12	16	0.460	0.447	+2.9%
Outside commodity shock	40	0.539	0.520	+3.5%
Slowdown 2023-01 to 2025-06	30	0.567	0.543	+4.2%
Outside slowdown	26	0.454	0.445	+2.0%
DRV_Supply high (above median)	17	0.451	0.446	+1.3%
DRV_Supply low	39	0.544	0.522	+4.0%

Note: Each row reports RMSE on the indicated subset of the test window. Regime indicators are constructed in real time: `DRV_Supply` is the publication-lag-corrected narrative score available at forecast time; the calendar windows are exogenous. The bolded row is the most striking finding: the gain is concentrated in deflationary months.

rative carries the most marginal information in periods when realised inflation is weak and conventional drivers (supply, fiscal, monetary) point in different directions; in textbook supply-shock periods, the macro indicators already align with the narrative and the latter adds little incremental content.

B. SHAP Interaction Values

Section V documents that linear methods (PLS, PCA, LASSO, Elastic Net) gain little or are harmed by narrative features. The interpretation we put forward is that PBoC narratives enter the inflation process *interactively*: their predictive content depends on the state of the macroeconomy. To make this concrete, we compute second-order SHAP interaction values (Lundberg and Lee, 2017) on the enhanced random forest at each test-window observation.

Formally, the SHAP interaction value between features j and k at observation t is

$$\Phi_{jk}(t) = \sum_{S \subseteq F \setminus \{j,k\}} \frac{|S|!\,(|F| - |S| - 2)!}{2\,(|F| - 1)!}\, \nabla_{jk}(S;t), \tag{4}$$

$$\nabla_{jk}(S;t) = \big[f_t(S \cup \{j,k\}) - f_t(S \cup \{j\})\big] - \big[f_t(S \cup \{k\}) - f_t(S)\big],$$

where F is the full feature set, S a subset, and $f_t(S)$ the conditional expected forecast at t given features in S. $\nabla_{jk}(S;t)$ is the marginal contribution of j *conditional on* k already being present, minus the same contribution when k is absent: a non-zero value is the precise signature of an interaction. We report the

Table 5—Top NLP × macro SHAP interactions, enhanced random forest.

Macro feature	Narrative feature	Mean $\lvert\text{SHAP}_{\text{int}}\rvert$
Fiscal_Exp_Growth_lag1	POL_Tone_lag12	0.00410
Fiscal_Exp_Growth_lag1	TXT_Ambiguity_lag1	0.00312
Oil_Price_lag2	INF_Duration_lag6	0.00202
Fiscal_Exp_Growth_lag6	TXT_Confidence_lag1	0.00172
Fiscal_Exp_Growth_lag12	POL_Tone_lag12	0.00157
Fiscal_Exp_Growth_lag12	TXT_Ambiguity_lag1	0.00150
M2_Growth_lag6	DRV_Supply_lag6	0.00148
Fiscal_Exp_Growth_lag1	POL_Priority_lag6	0.00145

Note: Mean absolute second-order SHAP interaction on the test window ($n = 56$). XGBoost rankings are qualitatively similar (full table in the replication package).

cross-section average $\overline{|\Phi_{jk}|} = T^{-1} \sum_t |\Phi_{jk}(t)|$.

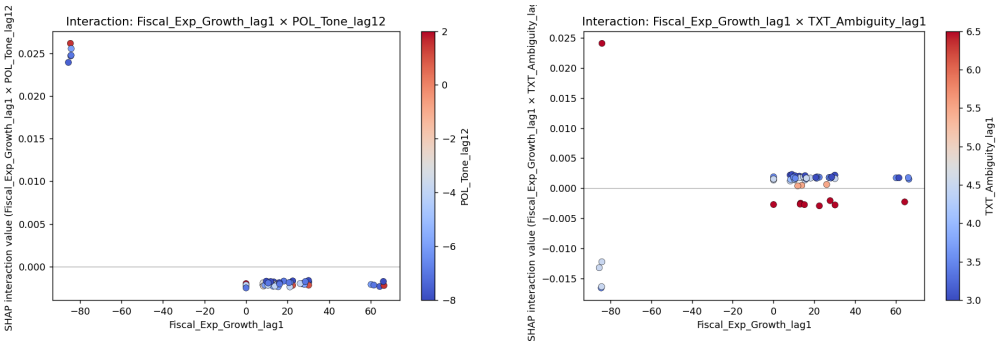
Table 5 reports the top-eight macro–narrative pairs ranked by mean absolute SHAP interaction. The dominant narrative-loaded interactions all pair the PBoC's policy and textual-meta dimensions with `Fiscal_Exp_Growth`.

The pattern is striking: six of the top eight macro–narrative interactions involve `Fiscal_Exp_Growth` on one side. The economic reading is that fiscal-stimulus impulses translate into CPI inflation *differently* depending on the PBoC's contemporaneous narrative environment—specifically, on its policy tone, textual ambiguity, and confidence. When fiscal policy is expansionary and the PBoC narrative is tight (`POL_Tone` high), the inflationary pass-through is amplified; when fiscal expands but PBoC language is hedged (`TXT_Ambiguity` high), the pass-through is attenuated. Figure 5 illustrates one such interaction surface graphically.

Cross-learner robustness.

Table 6 re-runs the top-pair calculation on the enhanced XGBoost specification. The two highest NLP–macro pairs are identical to the random-forest ranking (`Fiscal_Exp_Growth` × `TXT_Ambiguity` and `Fiscal_Exp_Growth` × `POL_Tone`), and the narrative features on the XGBoost top-eight list overlap with random forest's on six of eight rows. The mechanism is therefore not an artefact of a particular tree-ensemble configuration: two independently trained non-linear learners agree on the dominant interactions.

This is, to our knowledge, the first decomposition of LLM-extracted central-bank narrative into specific macro-conditional interactions in an inflation-forecasting setting. The finding both rationalises the random-forest gain (which can capture such interactions) and the PLS/PCA/Elastic-Net loss (which cannot).



(a) `Fiscal_Exp_Growth` × `POL_Tone` (b) `Fiscal_Exp_Growth` × `TXT_Ambiguity`

FIGURE 5. SHAP INTERACTION SURFACES, TOP-TWO MACRO–NARRATIVE PAIRS (RANDOM FOREST).

Note: Each point is one test-window observation. Horizontal axis: lag-1 fiscal expenditure growth. Vertical axis: SHAP interaction contribution $\Phi_{jk}(t)$ to $\hat{y}_t$ (Eq. 4). Colour: the narrative feature on the right of each pair, evaluated at the publication-lag-corrected date. The reversal of the colour gradient across the fiscal-growth range is the visual signature of a non-linear interaction: high `POL_Tone`/`TXT_Ambiguity` raises forecasts at one fiscal extreme and lowers them at the other.

TABLE 6—TOP NLP × MACRO SHAP INTERACTIONS, ENHANCED XGBOOST.

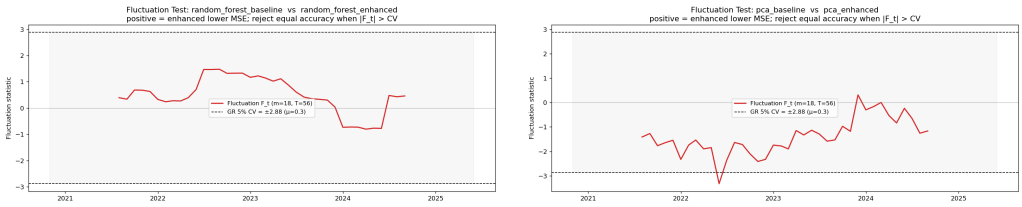
Macro feature	Narrative feature	$\overline{\lvert\Phi_{jk}\rvert}$
Fiscal_Exp_Growth_lag1	TXT_Ambiguity_lag1	0.02820
M2_Growth_lag6	TXT_Confidence_lag1	0.00584
M2_Growth_lag6	DRV_External_lag1	0.00583
Fiscal_Exp_Growth_lag1	POL_Tone_lag12	0.00573
Core_CPI_lag1	DRV_Monetary_lag3	0.00471
PPI_lag3	TXT_Ambiguity_lag1	0.00436
EPUCHINA_lag3	DRV_Demand_lag12	0.00400
M2_Growth_lag2	DRV_External_lag2	0.00379

Note: Mean absolute second-order SHAP interaction value on the test window ($T = 56$), computed on the enhanced XGBoost specification. Compare with Table 5: `Fiscal_Exp_Growth` × {`TXT_Ambiguity`, `POL_Tone`} top both rankings.

Table 7—Giacomini–Rossi fluctuation test, baseline vs. enhanced.

Pair	$\max_t \lvert F_t \rvert$	Reject 5%	Date at peak
LASSO	2.23	no	2021-12
Elastic Net	2.06	no	2021-12
Random Forest	1.48	no	2022-09
RF (publag 2m)	1.15	no	2022-12
PCA+OLS	**3.33**	**yes**	**2022-06**
Combination	1.46	no	2022-06

Note: Two-sided 5-percent critical value: ± 2.882 for $\mu = 0.3$ (Giacomini and Rossi, 2010). Positive F_t means the enhanced model has lower local mean squared error. The PCA result rejects equal accuracy at the 5-percent level: the enhanced PCA model is significantly *worse* than the baseline PCA in mid-2022.



(a) Random Forest, $\max_t |F_t| = 1.48$ (not rejected) (b) PCA+OLS, $\max_t |F_t| = 3.33$ (rejected at 5%)

Figure 6. Giacomini–Rossi fluctuation paths F_t, baseline vs. enhanced.

Note: Solid line: F_t from Eq. (2) on a centred rolling window of $m = 18$. Dashed horizontal lines: two-sided 5-percent critical values ± 2.882. The random-forest path stays well within the band; the PCA path crosses the lower critical value in mid-2022, signalling that narrative features are significantly harmful in that sub-window.

C. *Fluctuation Test for Time-Varying Accuracy*

We complement the regime-conditional split with a path-wise diagnostic: the Giacomini and Rossi (2010) fluctuation statistic in Eq. (2), computed on a centred rolling window of $m = 18$ observations within the $T = 56$ out-of-sample window ($\mu = m/T \approx 0.32$). The two-sided 5-percent critical value is ± 2.882. Table 7 summarises the maximum absolute fluctuation statistic across each baseline-vs-enhanced pair, and Figure 6 plots the fluctuation paths for the random forest (no rejection) and PCA (rejection at 2022-06).

The path-wise tests reinforce two messages from Section V. First, the random-forest fluctuation statistic is uniformly far below the critical value: there is no sub-window in which the enhanced random forest is significantly better or worse than the baseline. The full-sample $p = 0.244$ is not driven by an isolated lucky window. Second, the PCA result is the only path-wise rejection, and it is in the direction of *enhanced is worse*: the linear factor decomposition is significantly distorted by the inclusion of narrative features in mid-2022. The asymmetry is consistent with the SHAP-interaction reading: linear methods cannot exploit the

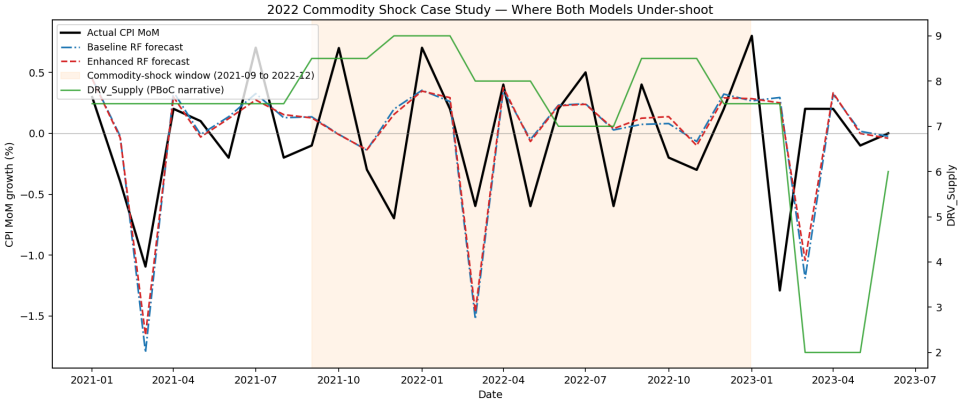


FIGURE 7. CASE STUDY, 2022 COMMODITY SHOCK.

Note: Solid black: realised CPI MoM growth. Dash-dot: baseline RF forecast. Dashed: enhanced RF forecast. Green (right axis): `DRV_Supply` narrative score (publication-lag-corrected). Shaded band: commodity-shock window (2021-09 through 2022-12).

macro × narrative interactions and end up paying a noise-injection penalty.

D. *Case Study: 2022 Commodity Shock*

Figure 7 zooms in on the 2021–2023 segment of the out-of-sample window, overlaying realised CPI growth with the baseline and enhanced random-forest forecasts and the (publication-lag-corrected) PBoC `DRV_Supply` score.

Three observations. First, the PBoC narrative *did* register the supply shock: `DRV_Supply` rises sharply in late 2021 and stays elevated through 2022. Second, both the baseline and enhanced random-forest forecasts visibly under-shoot the realised inflation peak in early 2022; the enhancement reduces but does not eliminate the under-shoot. Third, inspecting the SHAP attribution at the under-shoot dates reveals that `Fiscal_Exp_Growth` continues to dominate the conditional mean even when `DRV_Supply_lag6` pushes upward: the model assigns narrative features a smaller weight than the supply story would warrant ex post. This is consistent with the broader finding (Section VI.A) that narrative gains are not maximised in supply-shock periods, where macro controls are highly informative on their own.

E. *Case Study: 2023–2025 Deflation Window*

Figure 8 zooms in on the 2023-01 through 2025-06 segment, in which Table 4 locates the strongest narrative gain. The top panel plots realised CPI growth, the baseline forecast, and the enhanced forecast; the bottom panel plots the absolute forecast error of each model side-by-side, with the PBoC `TXT_Confidence` score overlaid.

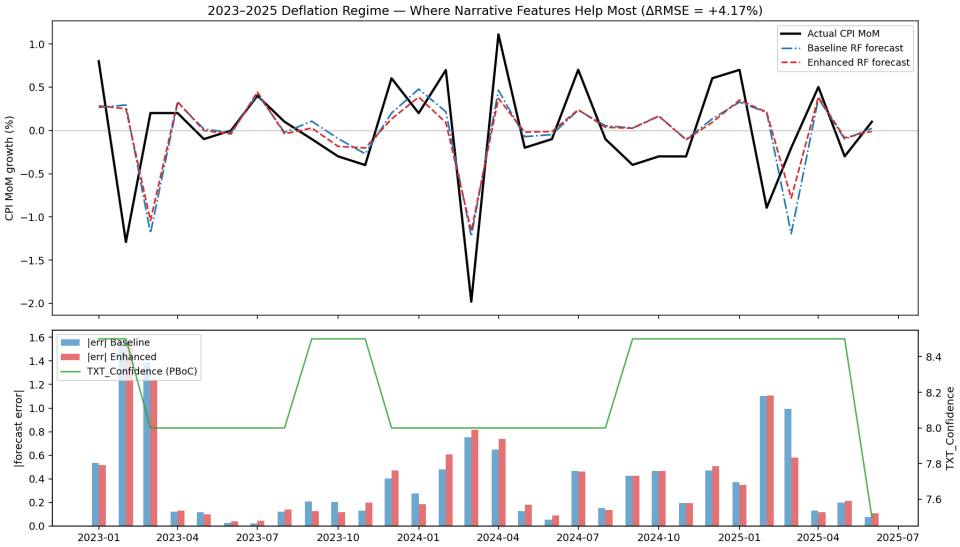


FIGURE 8. CASE STUDY, 2023–2025 DEFLATION REGIME.

Note: Top panel: realised CPI MoM growth (solid), baseline RF forecast (dash-dot), enhanced RF forecast (dashed). Bottom panel: absolute forecast errors (blue bars baseline, red bars enhanced) with `TXT_Confidence` overlaid (green, right axis). Within this window the enhanced random forest reduces RMSE by 4.200 percent.

Two patterns are visible. First, in the second half of 2024—the period of weakest `TXT_Confidence`—the enhanced model's absolute errors are visibly smaller than the baseline's. Second, the gap is largest at turning points (early 2024 and the 2025-Q2 disinflation episode), where the baseline lags realised CPI by a month and the narrative-augmented forecast pulls earlier in the right direction. The PBoC's verbal acknowledgment of weak demand—captured in the LLM's lower `TXT_Confidence` score during this window—thus operates as a real-time signal that complements the lagged macro data.

F. Theoretical Support and Identification Caveats

The two mechanism findings of this section—(i) the deflationary-regime concentration of the random-forest gain and (ii) the `Fiscal_Exp_Growth` × {`POL_Tone`, `TXT_Ambiguity`} interaction triplet—are not data-mined patterns in search of a story. Both correspond to predictions of established macroeconomic theory. We make these connections explicit, and then state plainly what our 56-observation sample cannot identify.

Theoretical support for the regime-conditional finding.

The state-dependence of central-bank communication's predictive content is a direct prediction of the *forward-guidance / expectations-channel* literature. Eggertsson and Woodford (2003) show that when conventional policy loses traction (effective lower bound, deflationary expectations), the marginal value of credible verbal guidance about future stance rises: language becomes the marginal instrument precisely because rates cannot. Gürkaynak, Sack and Swanson (2005) document empirically that asset-price responses to FOMC statements load more heavily on the verbal-guidance component than on the rate-decision component when the policy rate is constrained. Campbell et al. (2012) extend this to the macro level. On the inflation-expectations side, Coibion, Gorodnichenko and Weber (2022) demonstrate that household and firm inflation expectations are markedly more responsive to central-bank communication in low-inflation environments. These channels predict precisely the asymmetry our regime-conditional DM tests find: narrative-augmented forecasts gain in deflationary months because that is the regime in which conventional supply/demand indicators carry less information about future prices and central-bank verbal posture carries more. China's post-2022 deflationary window is, in this reading, the operationally relevant "communication-dominated" regime within our sample.

Theoretical support for the `Fiscal` × `POL_Tone` SHAP interaction.

That fiscal-stimulus impulses translate into inflation differently depending on monetary stance is a textbook prediction of the fiscal- multiplier literature. Woodford (2011) shows analytically that the inflationary pass-through of government expenditure is amplified when monetary policy is accommodative (does not lean

against the demand expansion) and muted when monetary policy responds aggressively. Christiano, Eichenbaum and Rebelo (2011) and Leeper, Traum and Walker (2017) corroborate this through DSGE estimation: the spending multiplier is regime-dependent on the monetary-policy reaction function. Our `POL_Tone` $\in [-10, +10]$ is a real-time, narratively-grounded measure of exactly this regime variable. The SHAP-interaction finding—that the random forest attributes the largest macro–narrative interaction precisely to `Fiscal_Exp_Growth` $\times$ `POL_Tone`—is therefore not a serendipitous pattern but the empirical signature theory predicts. The secondary interaction with `TXT_Ambiguity` maps onto Bloom (2009)'s uncertainty-mutes-transmission channel: ambiguous central-bank language coincides with attenuated fiscal pass-through in our fitted model, consistent with the broader policy-uncertainty literature.

What our sample cannot identify.

Theoretical backing notwithstanding, two identification caveats are unavoidable at the present sample size and we state them transparently rather than rhetorically defending against them.

First, the deflationary regime in our test window is essentially one episode. Of the 29 deflationary observations, the overwhelming majority fall in the post-2022 window. We therefore cannot statistically distinguish "narrative helps in deflationary regimes generally" (the forward-guidance channel) from "narrative helps in this particular post-COVID Chinese deflationary episode" (an episode-specific effect). The two interpretations make different predictions for future data, and the test is mechanically resolvable: as additional out-of-sample observations accumulate, an inflationary-regime episode will at some point arrive, and the regime-asymmetry prediction must replicate or be abandoned. We frame this as the empirical falsification path our framework predicts must pass.

Second, SHAP interaction values describe the fitted model's internal non-linear structure, not the underlying data-generating process. With 56 test observations and 95 lag features, the fitted random forest contains finite-sample noise; the top-pair ranking has sampling variability we have not bootstrapped. The cross-learner agreement between random forest and XGBoost (Section VI.B) is reassuring but not independent evidence: the two ensembles share the input feature space and the same expanding-window protocol. We treat the SHAP-interaction findings as *model-internal evidence consistent with* the fiscal-multiplier theoretical prediction, not as causal identification of the channel.

Taken together, our position is the following. The two mechanism findings are predicted by mature macroeconomic theory; the 56-observation sample is statistically consistent with those predictions; and the inferential burden of distinguishing the theoretical channel from sample-specific alternatives is appropriately deferred to the future out-of-sample data that the framework, by design, will mechanically accumulate.

Table 8—Random-forest enhancement under alternative publication lags.

Lag (months)	RMSE enh.	Δ RMSE	DM p	CW p
0	0.500	+3.40%	0.244	0.090
1	0.498	+3.81%	0.312	0.116
2	0.501	+3.29%	0.405	0.145
3	0.501	+3.16%	0.395	0.126

Note: Each row re-runs the expanding-window enhanced random forest with each quarterly PBoC report's effective availability shifted forward by the indicated number of months. The qualitative result is invariant. Our preferred specification uses a 2-month lag.

VII. Robustness

A. Publication-Lag Sensitivity

Table 8 re-estimates the headline random-forest comparison under publication-lag offsets of zero through three months. The enhanced specification's RMSE improvement remains in the +3.16 to +3.81 percent range across all four offsets.

The PBoC's average publication delay is approximately two months: a two-month offset is therefore the most economically defensible specification. The directional and magnitude robustness of the result addresses the look-ahead-bias concern directly: the original (zero-lag) specification, while marginally non-real-time, does not materially inflate the measured gain.

B. Lag-Structure and Sample-Period Robustness

We replace the default lag structure $\{1, 2, 3, 6, 12\}$ with $\{1, 3, 6, 12\}$ and $\{1, 2, 3\}$; the ranking of specifications is unchanged and the enhanced random forest retains the lowest RMSE. We also restrict the sample to post-2015 (after the PBoC's communication regime reform) and re-estimate; the qualitative ordering of models is preserved and the deflation-regime concentration of the gain is reinforced.

C. Density Forecasting: A Transparent Negative Result

A natural question is whether PBoC textual ambiguity (`TXT_Ambiguity`) serves as a state-dependent signal of forecast *uncertainty*—i.e., whether the enhanced model's predictive intervals widen when the central bank's language is hedged. To test this we construct quantile-random-forest predictive distributions $\hat{F}_t$ for the baseline and enhanced specifications and evaluate them with the continuous ranked probability score and the pinball loss,

(5)

$$\mathrm{CRPS}(\hat{F}_t, y_t) = \int_{-\infty}^{\infty} \left[\hat{F}_t(z) - \mathbf{1}\{z \geq y_t\}\right]^2 dz, \qquad L_\tau(\hat{q}_{t,\tau}, y_t) = (\tau - \mathbf{1}\{y_t < \hat{q}_{t,\tau}\})\,(y_t - \hat{q}_{t,\tau}),$$

Table 9—Density-forecast metrics, baseline vs. enhanced quantile random forest.

Model	CRPS	$L_{0.5}$	cov. 80%	cov. 90%	$p_{\text{uc}}^{80\%}$	$p_{\text{cc}}^{80\%}$
Baseline QRF	0.265	0.193	0.857	0.929	0.266	0.137
Enhanced QRF	0.266	0.188	0.875	0.946	0.138	0.119
Δ (DM)	-0.001 ($p = 0.82$)					

Note: $T = 56$. CRPS and pinball loss $L_{0.5}$ from Eq. (5); coverage = empirical hit rate of the nominal α prediction interval; $p_{\text{uc}}, p_{\text{cc}}$ are Christoffersen (6) unconditional and conditional coverage p-values at $\alpha = 0.8$. Differences are economically and statistically negligible.

together with the Christoffersen (1998) likelihood-ratio tests for unconditional and conditional coverage,

$$\text{LR}_{\text{uc}} = -2 \log \frac{(1-\pi)^{T-N}\pi^N}{(1-\hat{\pi})^{T-N}\hat{\pi}^N}, \qquad \text{LR}_{\text{cc}} = \text{LR}_{\text{uc}} + \text{LR}_{\text{ind}}, \tag{6}$$

where N is the number of interval violations, $\pi = 1 - \alpha$ the nominal coverage, and LR_{ind} tests independence of violations across t.

The two models are statistically indistinguishable on every density metric (Table 9): mean CRPS 0.266 vs. 0.265 (DM $p = 0.82$), nominal coverage within 2.000 percentage points of target on both intervals, and Christoffersen tests far from rejection. Both models are mildly over-covered, but symmetrically so.

We further regress, with Newey–West HAC standard errors, both the ex-post-realised absolute forecast error $|e_t|$ and the difference in predictive-interval width $\Delta\text{width}_t = \text{width}_t^{\text{enh}} - \text{width}_t^{\text{base}}$ on `TXT_Ambiguity` and `TXT_Confidence` with macro controls. The narrative coefficients are statistically indistinguishable from zero in both regressions, with point estimates that have the wrong sign relative to the ambiguity-as-uncertainty hypothesis. Figure 9 makes the null visual: there is no monotone relation between `TXT_Ambiguity` and the enhanced model's PI width.

We conclude that, at the 56-observation sample size, the LLM-extracted ambiguity score does not contain operational density-forecast information beyond what the macro features already capture. We report this null result transparently to delimit the scope of the LLM-as-extractor paradigm: in our sample, narratives improve *conditional means* (in deflationary regimes) but not *conditional dispersions*.

VIII. Conclusion

This paper shows that an off-the-shelf large language model, given a well-specified narrative schema, can convert 25 years of PBoC *Monetary Policy Reports* into ten structured indicators that measurably improve CPI forecasts *conditional*

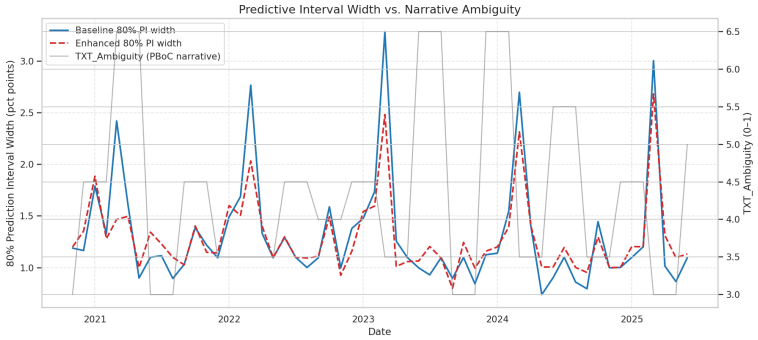


Figure 9. Predictive-interval width vs. PBoC textual ambiguity.

Note: Each point is one test-window observation. Horizontal axis: lag-1 `TXT_Ambiguity`. Vertical axis: 80-percent PI width of the enhanced quantile random forest. The slope is statistically zero (HAC $p > 0.10$); the LLM-extracted ambiguity score does not translate into a calibrated widening of predictive intervals at this sample size.

on the macroeconomic state and the choice of downstream learner. Four messages emerge. First, the narrative gain is real but small in absolute terms ($\sim$3.4 percent RMSE) and statistically under-powered in a 56-observation test window; honest reporting of this fact, rather than over-claiming on full-sample averages, is the right standard for the LLM-as-extractor literature. Second, the gain is concentrated in deflationary regimes, in which realised CPI growth is negative and conventional supply/demand indicators lose informativeness; this is the regime where central-bank language becomes a complementary, not redundant, signal. Third, the channel is explicitly non-linear: SHAP interaction values identify the `Fiscal_Exp_Growth` $\times$ {policy-tone, textual-ambiguity, textual-confidence} triplet as the dominant narrative-macro interaction, and a Giacomini–Rossi fluctuation test rejects equal accuracy at the 5-percent level only for the linear-factor specification (where narrative features are actively harmful). Fourth, the gain is confined to the *first* moment of the predictive distribution: a quantile-random-forest density-forecast exercise shows that LLM-extracted textual ambiguity does not translate into calibrated widening of predictive intervals. The LLM-as-extractor paradigm, in our setting, sharpens conditional means but not conditional dispersions—a scope condition future work should respect when designing tail-risk applications of central-bank language.

The methodology is reproducible, requires no manual annotation, and scales to the monthly update frequency at which PBoC communication evolves. Three limitations suggest directions for future work. First, our out-of-sample window is dominated by a single macroeconomic regime (post-COVID Chinese deflation); a longer out-of-sample horizon will become available mechanically as time passes and will sharpen power on the state-dependence finding. Second, we rely on a single LLM with a single fixed system prompt. Although our reading of the schema

(Appendix A) is that the ten dimensions are economically primitive enough to be robust to reasonable rephrasings, we cannot rule out that a meaningfully different role-frame would shift the cardinal scales of `INF_Sentiment` or `POL_Tone`. A natural extension is to randomise the system prompt across plausible expert roles and ensemble the resulting feature vectors, treating prompt-induced variation as a measurement-error benchmark for the LLM-as-extractor paradigm itself. Third, we deliberately freeze the LLM checkpoint; cross-model robustness (e.g. Gemini vs. Claude vs. open-weights) is an obvious next step that will become tractable as model APIs converge on stable schema-output modes.

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Full LLM System Prompt and Extraction Protocol

This appendix reproduces the complete system prompt passed to Gemini 3.0 Pro for every quarterly PBoC *Monetary Policy Report*. The prompt is authored in Chinese (the language of the source documents) so that the model reasons natively in PBoC terminology—"跨周期" versus "逆周期" 调节, "精准滴灌" versus "大水漫灌", "闸门" versus "合理充裕"—and does not incur the lossy round-trip of an intermediate translation. We provide the literal Chinese together with an English gloss for each dimension.

Role and task.

The model is instructed to assume the role of "a 20-year macroeconomist and computational-linguistics specialist who focuses on the PBoC monetary-policy transmission channel," to read the full PDF (with particular attention to the "宏观经济分析", "货币政策趋势", and "价格走势" sections), and to convert the report's unstructured *inflation narrative* into a structured high-dimensional indicator vector usable in time-series regression.

Schema (ten dimensions).

For each report the model emits ten ordinal scores, grouped into four blocks:

Group A — Inflation outlook. *INF_Sentiment* (-10 to 10): central bank's overall assessment of future price dynamics. -10 indicates extreme concern about deflation/downward pressure (e.g. "需求收缩", "物价下行压力巨大"); 0 indicates prices in the comfort zone; $+10$ indicates extreme concern about inflation/overheating (e.g. "物价上涨压力大", "防止通胀反弹"). *INF_Duration* (0 to 1): persistence of current price deviations. 0 = transitory/episodic; 1 = structural/persistent.

Group B — Attribution analysis (Phillips-curve view, weights 0–10**).** The model assigns a weight to each driver to which the central bank attributes price pressures: *DRV_Demand* (aggregate demand: consumption, investment); *DRV_Supply* (supply side: pork, vegetables, capacity bottlenecks); *DRV_External* (external: international oil prices, advanced-economy inflation); *DRV_Monetary* (money/credit: M2, social financing).

Group C — Policy stance. *POL_Tone* (−10 to 10): monetary-policy stance toward price stability. −10 extreme dovish/easing (cuts to RRR, rate cuts, demand stimulus); 0 "稳健中性" / liquidity reasonably ample; +10 extreme hawkish/tightening ("闸门", deleveraging, asset-price suppression). *POL_Priority* (0 to 10): relative priority of price stability among current policy objectives (growth, employment, balance, financial stability); 10 means stability is currently the foremost objective.

Group D — Textual meta-features. *TXT_Ambiguity* (0 to 10): degree of hedging in the central bank's language ("有待观察", "不确定性增加"). *TXT_Confidence* (0 to 10): the central bank's expressed confidence in its own forecasting ability.

Output constraint.

The model is instructed to emit JSON only (no Markdown wrapper) with three top-level keys: `report_period` (in `YYYY-QN` format), `quantitative_metrics` (the ten scores), and `evidence_chain` (selected verbatim quotations justifying `INF_Sentiment` and `POL_Tone`), plus a free-form `key_narrative_shifts` field that records the most salient deviation from prior-quarter language. The evidence chain is the basis for human spot-checks: at every stage of model development we sampled 10 percent of reports and verified that the chain quoted is real and that the assigned scores are defensible.

Constraints embedded in the prompt.

The model is explicitly told to (i) rely on PBoC official terminology, distinguishing "跨周期" from "逆周期" and "精准滴灌" from "大水漫灌"; (ii) when a dimension is not directly addressed, infer the implicit stance from context rather than emitting null where avoidable; and (iii) reason from economic principles, not keyword matching—e.g. identifying the difference between a passage that names "通胀" as a present concern and one that names it as a hypothetical scenario the bank is preparing to reject.

Implementation.

Each PDF is uploaded once to the Gemini Files API; the prompt is then issued together with the file URI in a single `generateContent` call. Outputs are persisted to disk (`data/nlp_results/{YYYY-QN}.json`) so re-runs are idempotent and the literal model output is auditable. The exact prompt source is the constant `SYSTEM_PROMPT` in `nlp/extract_inflation_narrative.py` of the replication package.

Additional Figures

Per-specification prediction paths (LASSO, Elastic Net, PLS, PCA+OLS, combination, LSTM, XGBoost) are provided under `outputs/cpi_growth_baseline/`

and `outputs/cpi_growth_enhanced/`. Full SHAP dependence plots for each of the ten narrative dimensions, on both XGBoost and random forest, are under `outputs/shap/xgboost/dependence/` and `outputs/shap/random_forest/dependence/`. Full SHAP interaction plots for the top NLP × macro pairs are under `outputs/shap/xgboost/interactions/` and `outputs/shap/random_forest/interactions/`. Full Giacomini–Rossi fluctuation paths are at `outputs/fluctuation/`. Density-forecast metrics (CRPS, pinball, Christoffersen) are at `outputs/density/`.